

Daisy



File: Seed3-DevKit-Desktop\_1.kicad\_sch

Peripherals



File: Seed3-DevKit-Desktop\_2.kicad\_sch

Audio



File: audio.kicad\_sch

POWER

Selecting R2 with formula:  
 $R2 = (V_{out}/V_{ref} - 1) \times R1$   
Where  $V_{ref} = 1.255V$   
For  $V_{out}$ : where  $R1 = 10K$  and  $R2 = 62K$   
 $(R2/R1)+1 \times V_{ref} = V_{out}$   
 $V_{out} = 9.036V$   
(measured: 8.919V on hardware)

Jump Point  
In case of troubleshooting  
or for testing power via Battery if we want

This will drop a fair amount of voltage  
Measured at 8.80V on hardware

This will drop a fair amount of voltage  
Measured at 7.975V on hardware

The +5V is powering:  
- Headphone Amp  
- DAC output op-amp  
- Reference voltage for CV inputs  
- 6002 buffer for AREF\_AUDIO\_BIAS

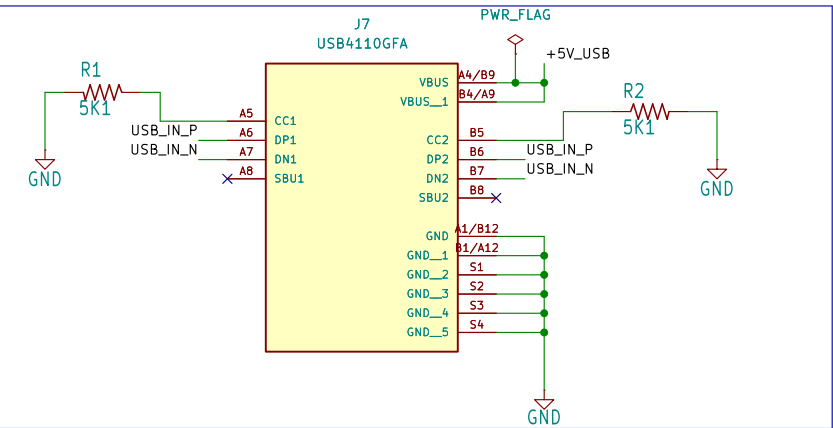
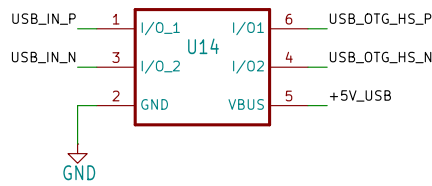


QR1

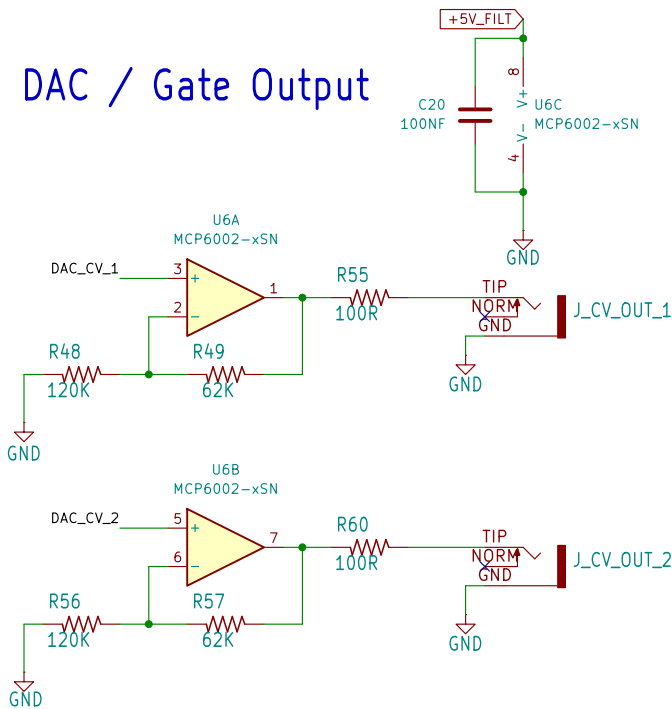


- H1 MOUNT-PAD-ROUND3.5
- H2 MOUNT-PAD-ROUND3.5
- H3 MOUNT-PAD-ROUND3.5
- H4 MOUNT-PAD-ROUND3.5

USB C

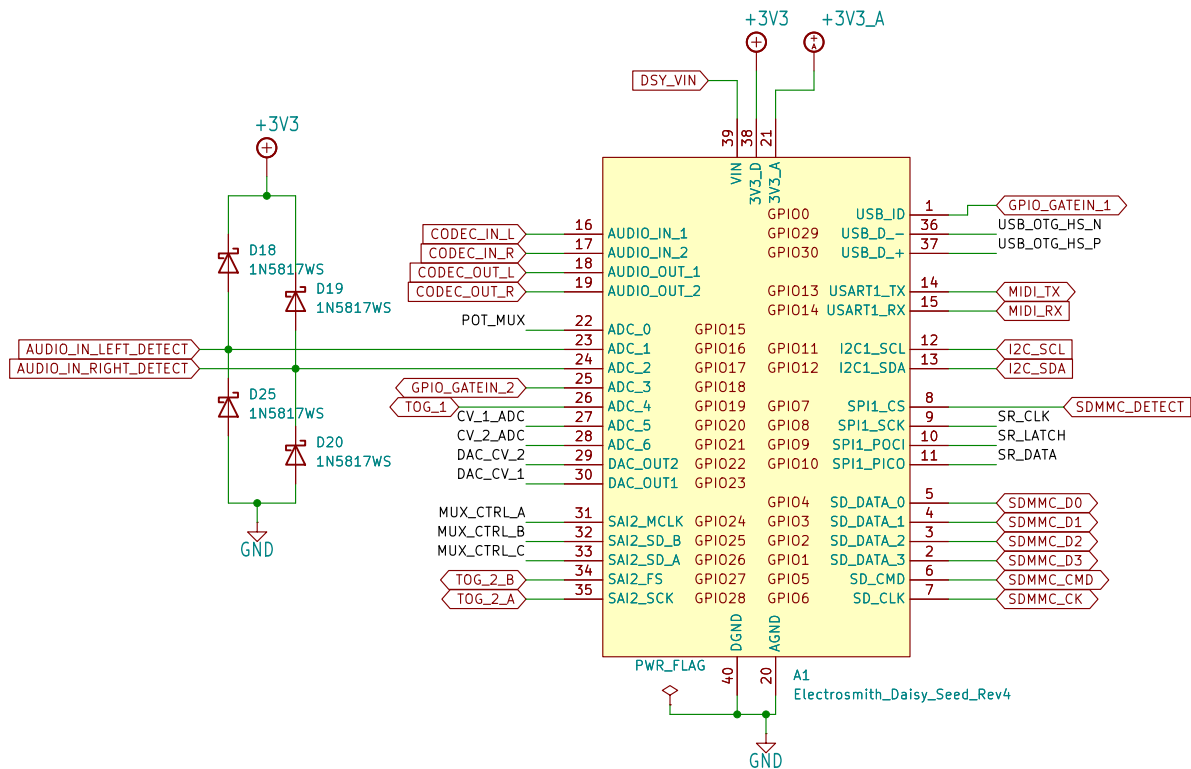
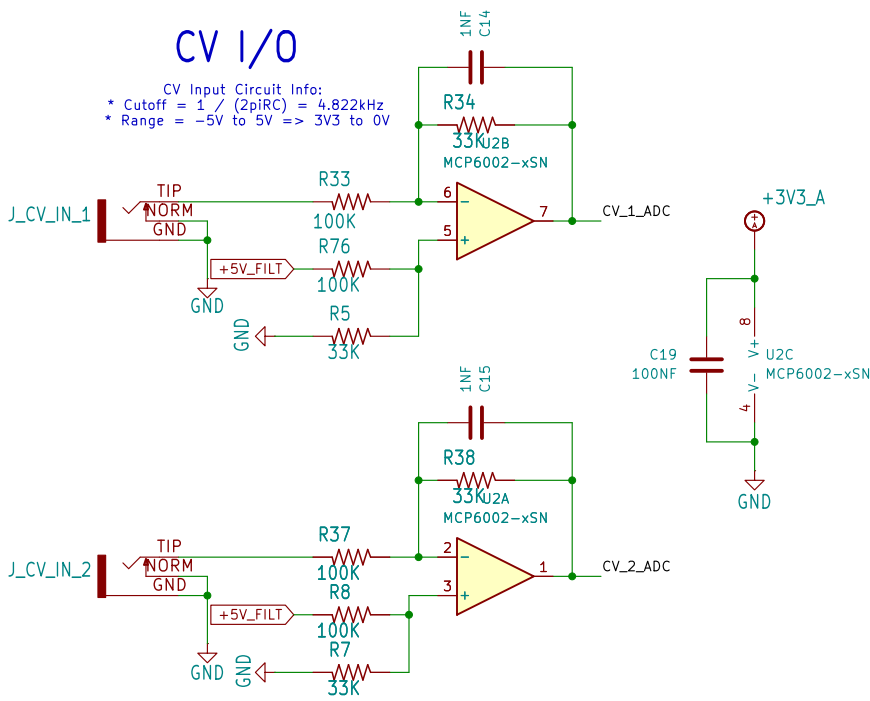


DAC / Gate Output

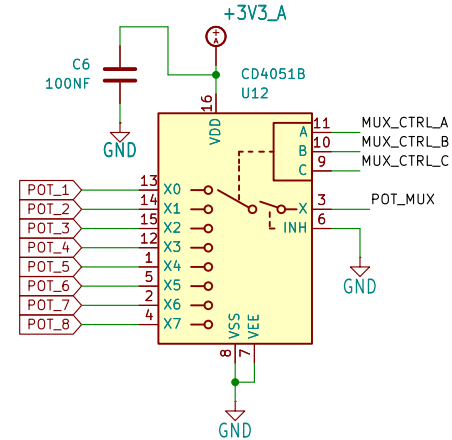
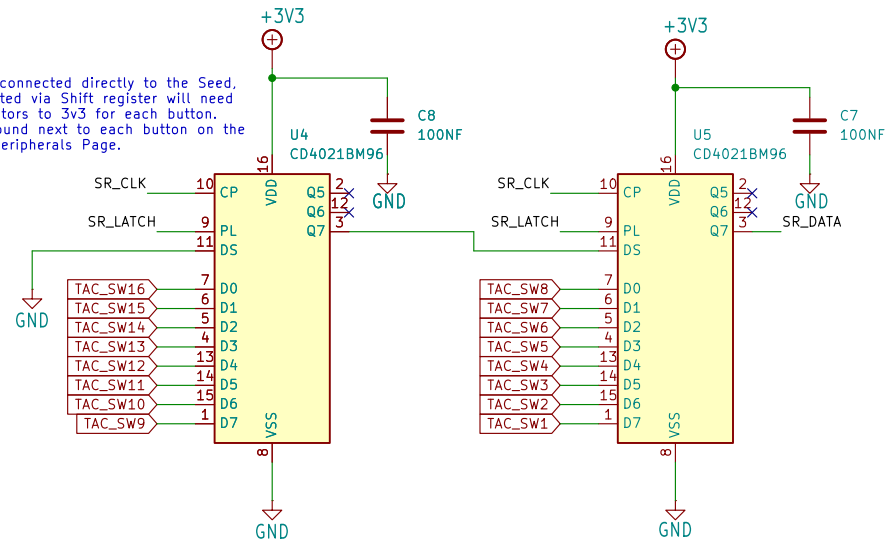


CV I/O

CV Input Circuit Info:  
\* Cutoff =  $1 / (2\pi RC) = 4.822kHz$   
\* Range = -5V to 5V => 3V3 to 0V



Unlike buttons connected directly to the Seed,  
buttons connected via Shift register will need  
pull up resistors to 3v3 for each button.  
These can be found next to each button on the  
Peripherals Page.



# LEDs

The image displays 16 individual LED circuit diagrams arranged in a 4x4 grid. Each diagram shows an LED connected in series with a 2K resistor to a positive supply, with the other end of the LED connected to ground. The LEDs are labeled D1 through D16, and the resistors are labeled R43 through R72. The LED labels are in blue, and the resistor labels are in red.

| LED Label | Resistor Label | Resistor Value |
|-----------|----------------|----------------|
| D1 LED    | R43            | 2K             |
| D2 LED    | R35            | 2K             |
| D3 LED    | R36            | 2K             |
| D4 LED    | R39            | 2K             |
| D5 LED    | R40            | 2K             |
| D6 LED    | R41            | 2K             |
| D7 LED    | R42            | 2K             |
| D8 LED    | R44            | 2K             |
| D9 LED    | R45            | 2K             |
| D10 LED   | R46            | 2K             |
| D11 LED   | R47            | 2K             |
| D12 LED   | R58            | 2K             |
| D13 LED   | R69            | 2K             |
| D14 LED   | R70            | 2K             |
| D15 LED   | R71            | 2K             |
| D16 LED   | R72            | 2K             |

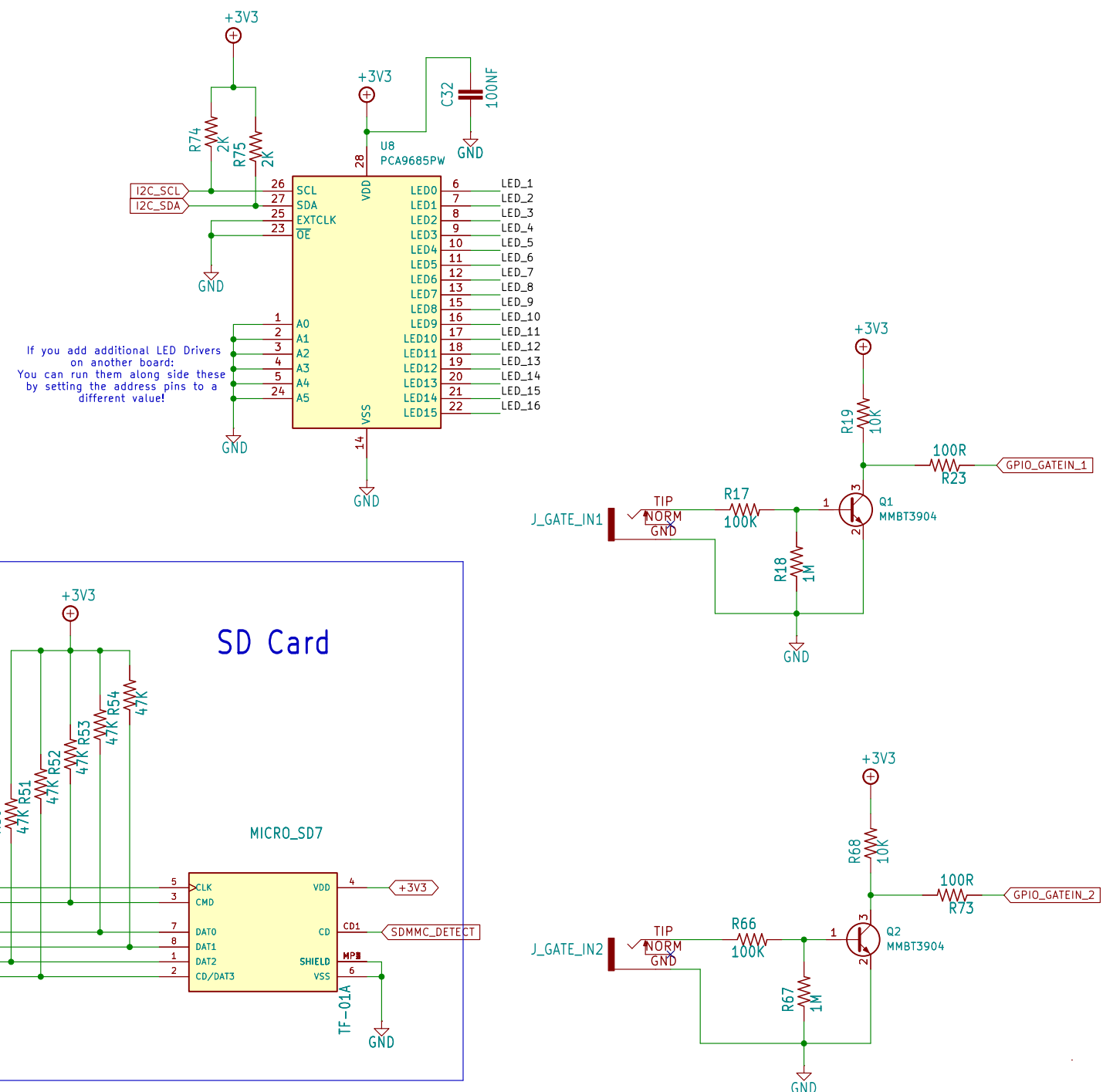
# MIDI

The top diagram shows a MIDI IN connection. It features a 1N5817WS diode (D21) in series with a 220R resistor (R4) connected to MIDI\_IN\_4. The other end of R4 is connected to pin 1 of an H11L1 optoisolator (U11). Pin 2 of U11 is connected to MIDI\_IN\_5, and pin 5 is connected to GND. Pin 6 of U11 is connected to a +3V3 supply. Pin 4 of U11 is connected to a 270R resistor (R3) leading to +3V3, and a 100NF capacitor (C1) leading to GND. Pin 3 of U11 is connected to MIDI\_RX.

The bottom left diagram shows a MIDI IN connection to a J10 connector. MIDI\_IN\_5 is connected to pin 1 of J10, and MIDI\_IN\_4 is connected to pin 4 of J10. Pin 2 of J10 is connected to GND. Pin 3 of J10 is connected to a 100NF capacitor (C34) to GND and a 100NF capacitor (C47) to a PWR\_FLAG output. Pin 5 of J10 is connected to GND.

The bottom right diagram shows MIDI OUT and MIDI THRU connections. The MIDI OUT section uses a J11 connector with MIDI\_TX connected to pin 1, pin 4 to GND, and pin 2 to a 10R resistor (R59) leading to MIDI\_TX. A +3V3 supply is connected to pin 3 of J11 through a 10R resistor (R6). The MIDI THRU section uses a J8 connector with MIDI\_RX connected to pin 1, pin 4 to GND, and pin 2 to a 10R resistor (R29) leading to MIDI\_RX. A +3V3 supply is connected to pin 3 of J8 through a 10R resistor (R10).

The PWR\_FLAG output is noted as being awkward but recommended for input isolation.



# AUDIO I/O

